

FREDERICK DEHMEL

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EDUCATION

University of California, Berkeley

Aug 2023 – May 2027

B.A. Computer Science, B.A. Pure Mathematics

Technical GPA: 3.94 / 4.0

Coursework: Data Structures (A+), Efficient Algorithms, Computability & Complexity (A+), Discrete Mathematics & Probability, Computer and Network Security, Signals and Systems, Deep Neural Networks, Multivariable Calculus, Linear Algebra & Differential Equations (A+), Honors Linear Algebra, Honors Abstract Algebra, Quantum Computing (A+)

Graduate Coursework: Algebra (Groups, Rings and Fields), Cryptography, Proving TCS and Math Theorems in Lean, Quantum Algorithms, Randomness and Computation

SKILLS

Tools: PostgreSQL, MongoDB, PyTorch, TensorFlow, FastAPI, React, Docker, Kubernetes, Pandas, Spark, Git

Languages: Python, Java, C/C++, SQL, Go, JavaScript, Rust, R, MATLAB, Assembly, RISC-V, Shell/Bash, HTML, Lean4

EXPERIENCE

Research Intern — Cryptography Group | Amazon Web Services, New York, NY

Jun 2026 – Aug 2026

- Joining the AWS cryptography research group (Fabrice Benhamouda, Shai Halevi, Hugo Krawczyk, Yuval Ishai, Tal Rabin) to conduct research on secure multi-party computation (MPC) and fully homomorphic encryption (FHE), with applications to encrypted search.
- Contributing to the design and analysis of cryptographic constructions, empirical evaluation of cryptanalytic attack complexity, and integration of primitives into proof-of-concept systems; implementation in Rust/C++.

Summer Fellow | YCombinator

Jun 2026 – Aug 2026

- Selected as one of approximately 50 students for the YCombinator Summer Fellows program.
- Awarded approximately \$325,000 in cloud and AI compute credits from leading infrastructure providers to accelerate product development.

Head Teaching Assistant — CS 161 (Computer Security) | UC Berkeley EECS

Aug 2025 – Present

- Directed 20 TAs supporting 500+ students, managed administrative logistics and staff scheduling. Led up the exam-writing, writing the midterm and final examinations.
- Received the **2026 Outstanding GSI Award**, given to the **top 9%** of UC Berkeley (u)GSIs.

Volunteer Instructor — San Quentin | Prison University Project

Jun 2026 – Present

- Teach computer science and mathematics to incarcerated students at San Quentin Rehabilitation Center, leading lessons and providing individualized academic support.

Researcher — Pseudorandom Codes | UC Berkeley EECS

Aug 2025 – Present

- Conducting research under Professor Venkat Guruswami on pseudorandom error-correcting codes (PRCs), a cryptographic primitive with applications to watermarking generative AI
- Developing original proofs on new PRC constructions under alternative cryptographic assumptions (e.g., LPN, planted XOR) and analyzing adaptive robustness guarantees against non-oblivious error channels

Lecturer — CS 161 (Computer Security) | UC Berkeley EECS

May 2025 – Aug 2025

- Designed and delivered 20+ lectures on cryptography, network security, and secure software design to 150+ students.
- Earned $\frac{4.8}{5.0}$ average course evaluation rating for instructional clarity and content depth.

Researcher — Quantum Coding Theory & Formal Methods | UC Berkeley EECS

Aug 2025 – Present

- Developing a structured Lean 4 library to formalize algebraic constructions, verifying asymptotic bounds and complexity-theoretic reductions central to quantum error correction under professor Venkat Guruswami.
- Synthesizing concepts from quantum computing and finite fields to create a rigorous framework for verifying coding theory theorems, bridging theoretical abstractions with machine-checkable proofs.

ML Research Mentor / Principal Investigator (PI) | Algoverse AI Research

Mar 2025 – Sep 2025

- Mentored 10+ researchers through 15+ seminars on LLMs and MLOps; standardized Dockerized workflows across 5 tracks; guided 3 papers to conference submissions while coordinating with OpenAI and other involved companies.
- Delivered tailored educational curricula spanning transformer architectures, alignment, evaluation, and deployment; translated cutting-edge research into reproducible experiments and production-ready pipelines for early-career researchers.

PUBLICATIONS

A Symplectic Proof of the Quantum Singleton Bound

Frederick Dehmel, Shilun Li

Preprint in submission— 2026

[\[ArXiv\]](#)

PROJECTS

Secure File Sharing System

Golang | TLS | Socket Programming | OpenSSL | Pytest | Systems Design

- Built AES-CTR + RSA encrypted client with per-user key management; MAC verification blocked 93% of tampering across 1,000+ penetration tests with no integrity failures.
- Reduced transfer time by 40% through parallel chunked streaming; supported 5GB file sharing securely across 500+ validated trials.